COMP 10280
Programming I (Conversion)
Practical Sheet 19
Dé hAoine, 16 Deireadh Fómhair 2026

For each of the following questions, write an algorithm in pseudocode first before writing a Python program. Submit your algorithms in psuedocode as well as your Python programs.
When writing functions, use one-line or multi-line docstrings, as appropriate, to document your functions.

1. Write a Python function that takes a number in base 10 and a base, both positive `int`s and where the base is ≥ 2 and ≤ 10, and, using the algorithm presented in lectures, returns a string representing the number in the base supplied.

   Save this program as `p19p1.py`.

2. Write a function that takes a string of digits, representing a number, and a base (an `int` between 2 and 10) as arguments, and returns the number in decimal (Base 10). Your function may assume that only the digits appropriate to that base appear in the string, eg only "0"s and "1"s appear in a string representing a number in Base 2, only "0"s, "1"s and "2"s appear in a string representing a number in Base 3, etc.

   Save this program as `p19p2.py`.

3. Write a program that reads a file and checks that brackets (ie `(` and `)`, `<` and `>`, { and } and `[` and `]`) are balanced, ie there should never be a situation where there are more right brackets of a particular type than there are corresponding left brackets and the total number of right brackets should equal the number of left brackets (You do not need to consider the interleaving of different bracket types). Your program should return the total number of each bracket and a message indicating whether or not the file has balanced brackets.

   Save this program as `p19p3.py`.

Please upload your work to the Brightspace site before Sunday evening.

You should keep a copy of your programs for your portfolio.